EcoExplain-VLM · experimental workflow

Kanha Tiger Reserve · dry season T0 Jan–Feb 2022 / T1 Jan–Feb 2023

01 Multi-sensor inputs

Sentinel-2 optical
RGB + 10 m bands
NDVI · EVI · NDMI · NBR

Sentinel-1 SAR
VV / VH backscatter
descending pass

CHIRPS rainfall
Window totals
T0 and T1

ERA5-Land
Temperature & soil moisture
anomalies vs 2015–2021

NASADEM terrain
Elevation · slope
static context

02 Preprocessing

Earth Engine · matched composites, cloud mask, 640 m patches

Study rectangle 80.433–81.050°E, 22.117–22.450°N · 1,200 candidate centres · 15 GeoTIFFs per patch · T0/T1 RGB chips for the VLM

03 Labels and evidence

Paired RGB chips
True-colour T0 and T1 on a 512 × 512 canvas
image input (BLIP-2: one stitched image)

Automatic class (gold)
ΔNDVI percentiles of the 1,200-patch pool
q15 = −0.038 · q85 = +0.003

Structured evidence card
Indices, SAR, climate, terrain
last line repeats candidate_class=

04 Dataset freeze

366 patches → train 256 / development 55 / test 55 (seed 42)

Gold = candidate_class · majority baseline on test = 34.5% · same 55 test patches in every protocol below

05 Three evaluation tracks (inputs differ: do not mix the scores)

A Zero-shot classification
Qwen2-VL-2B, LLaVA-0.5B: T0/T1 + full card
PaliGemma-3B: T0/T1 + NDVI line only
BLIP-2: stitched T0|T1, no card
evidence_only: label line removed

B Class LoRA (16 patches, 1 epoch)
Qwen2-VL-2B · r = 8 · full card at train and test
first 16 training patches (9 / 6 / 1)
256-patch runs reached 100% by
reading the label, so they are excluded

C Explanations (label removed)
free-form prompt → structured cite-only prompt
claim directions checked against measured Δ
explanation LoRA pilot: 16 filtered
2B targets, all labelled decline